

HW06 — AI Audit Report

Student: Lê Tuấn Lộc (23127404)
AI usage declaration: **AI tools were used to assist test design, artifact generation, review, and documentation.**

Evidence basis and fidelity

This audit records AI-assisted work that is recoverable from version-controlled artifacts, Git history, and retained execution evidence. A platform-native export of every chat turn is not available. Accordingly, the report identifies the tool, time period, task request, retained output, and human review decision for each recoverable interaction; it does not invent missing prompts, timestamps, or model responses.

Field	Value
AI tool/model	Codex (GPT-5)
Work period	2026-08-23 to 2026-08-24 (UTC+07)
SUT	EShop backend at <code>http://127.0.0.1:3000</code>
Human responsibility	Select scope, approve contract oracles, audit generated cases, reproduce defects, create account-owned evidence, and approve the final diagrams and report.

Recoverable interaction register

R1 — Requirement review and scope

- **Date/evidence:** 2026-08-23; commit `ef27f9a` .
- **Task request:** Review the official HW06 specification and prepare a compliant API-testing scope.
- **AI output:** Selected three APIs from distinct pools: `POST /api/login` , `POST /api/checkout` , and `PUT /api/admin/orders/:id/status` .
- **Retained evidence:** `specs/2026.HW06.API Testing_En.pdf` , `specs/manual.md` , `specs/HW06_submission_checklist.md` , and commit `ef27f9a` .
- **Human decision:** Approved the three-API scope.

R2 — Test-case generation and audit matrix

- **Date/evidence:** 2026-08-23; commits `662c2b4` and `105610e` .
- **Task request:** Produce broad candidate API tests and retain a reviewable data-driven matrix.
- **AI output:** Created 35 candidate cases per API, Postman collections, iteration data, and audit fields: `source` , `audit_label` , `audit_reason` , `contract_status` , and `expected_status` .
- **Retained evidence:** `postman/data/*.json` , `postman/collections/*.json` , and `excel/23127404_HW06_API_Test_Cases.xlsx` .

- **Human decision:** Added five extension cases per API and separated the contract oracle from observed SUT behavior. Final total: 105 AI-generated and 15 human-extended cases.

R3 — Newman execution evidence

- **Date/evidence:** 2026-08-23T15:23:46.653Z.
- **Task request:** Execute the suite against the local SUT and preserve verifiable raw evidence rather than simulated screenshots.
- **AI output:** Implemented isolated fixture handling and an evidence-preserving runner.
- **Retained evidence:** `scripts/run_hw06_verified_evidence.js` , `evidence/execution-manifest.json` , and `newman-reports/` .
- **Observed result:** 120 primary cases, 320 HTTP requests, 688 assertions, and 0 runner failures.
- **Human decision:** Accepted raw Newman JSON as the source for numeric claims.

R4 — Defect analysis and reporting

- **Date/evidence:** 2026-08-23 to 2026-08-24; commits `dc362ec` and `272f1a6` .
- **Task request:** Investigate observed deviations and distinguish independent root causes from repeated manifestations.
- **AI output:** Consolidated the selected scope into six root-cause defects: login counter behavior, lock duration, plaintext password exposure, cart-rule bypass, missing admin-role authorization, and an invalid terminal-state transition.
- **Retained evidence:** `bugs/bug-report.md` , raw Newman evidence, GitHub Issues [#162–#167](#) , and `bugs/screenshots/` .
- **Human decision:** Created the Issues and captured authentic issue pages.

R5 — CI/CD evidence

- **Date/evidence:** 2026-08-23 to 2026-08-24; commits `57a8474` , `f15130e` , `d07a9b1` , and `aa32d91` .
- **Task request:** Run the API evidence suite in GitHub Actions on the working branch and retain both all-pass and intentional-failure evidence.
- **AI output:** Added the workflow and delivery copy at `.github/workflows/api-tests.yml` and `cicd/workflows/api-tests.yml` .
- **Retained evidence:** [all-pass run](#) , [intentional-failure run](#) , and `cicd/screenshots/` .
- **Human decision:** Confirmed both account-owned captures and restored the test expectation to `200` .

R6 — Agent-skill design and correction

- **Date/evidence:** 2026-08-23 to 2026-08-24; commits `f18b53c` and `03ca938` .
- **Task request:** Build a reusable API-test-generator skill for the selected APIs, then address execution defects found in review.
- **AI output:** Created the generator, skill instructions, pseudocode, Mermaid sources, and test-artifact synthesis. The reviewed implementation bootstraps real protected-suite tokens and binds admin requests to iteration-specific `orderId` values.
- **Retained evidence:** `agent-skills/api-test-generator/` , `agent-skills/diagrams/` , and commit `03ca938` .
- **Human decision:** Reviewed and approved the final Mermaid design sources and exported PNG diagrams.

R7 — Submission evidence and documentation

- **Date/evidence:** 2026-08-24; commits `272f1a6` , `3faa5ba` , and `bb9056d` .
- **Task request:** Link account-owned evidence and complete submission documentation.
- **AI output:** Connected GitHub Issues, CI captures, the Postman Console capture, and the Unlisted video to the delivery artifacts.
- **Retained evidence:** `README.md` , `report/23127404_HW06_API_Testing_Report.md/.pdf` , `cicd/` , `postman/screenshots/postman_console_23127404.png` , and [video demonstration](#).
- **Human decision:** Confirmed evidence authenticity, video availability, group scope, deadline, and final submission readiness.

Review conclusions

1. A `BUG DETECTED` result documents an observed defect reproduction; it is not a claim that the SUT conforms to its contract.
2. Stateful flows require controlled fixtures. Pool A isolates account state; Pool B exercises login, cart, checkout, and cart query; Pool C seeds each required order state.
3. Repeated manifestations are reported as six root causes rather than inflated into independent bugs.
4. GitHub Issue, CI, and Postman Console screenshots are account-owned tool captures, not generated images.
5. The final artifact set includes the public evidence links, Postman header capture, diagrams, AI audit, reports, and Unlisted video link.

Audit closure

All recoverable AI-assisted work is documented above. No additional platform-native chat export exists, and no missing interaction content has been reconstructed from memory.